

Monitoring Specification

Model Name: HC Manuscript Evaluator
Model ID: HC-QG-001
Tier: Tier 1 (High Risk)
Version: 4.0
Date: April 2026

1. Objective

This document defines the monitoring framework for the HC Manuscript Evaluator. It specifies the key risk and performance metrics to be tracked, the thresholds that trigger intervention, and the governance responses required when those thresholds are breached.

The objective is to provide continuous visibility into model health in production while ensuring full alignment with the Risk Control Matrix (RCM). Each monitored metric directly maps to a defined risk and functions as an ongoing validation mechanism.

2. Monitoring Philosophy

As a Tier 1 (High Risk) system, this model requires continuous verification. Monitoring ensures that assumptions validated pre-deployment (accuracy, fairness, legal compliance, and human oversight) remain valid in a live environment.

This framework emphasizes:

- Direct linkage between risks and metrics
- Clearly defined thresholds for intervention
- Explicit and actionable governance responses
- Independence from specific technical implementation tools

3. Monitored Metrics & Alerting Triggers

Metric ID	Metric Name & Linked Risk	Calculation Logic	Threshold & Severity	Response Protocol
M-01	Hallucination Rate (R-02)	Percentage of fabricated comparable book titles identified in a monthly audit	> 5% (CRITICAL)	Immediate model suspension. Notify MRM Lead and Technical Lead.

		sample		Conduct root cause analysis prior to reinstatement.
M-02	Bias Drift (R-01)	Difference in average scores between standard queries with and without author-identifying information, and calculation of disparate impact ratio (DIR)	Avg difference > 0.5 OR DIR outside 0.8-1.25 (WARNING)	Notify Editorial Director and MRM Lead. Initiate formal bias review and investigate recent outputs.
M-03	Automation Bias Indicator (R-04)	Percentage of cases where editor decisions significantly differ from AI recommendations (override rate)	< 30% or > 70% for 2 consecutive months (WARNING)	Notify Editorial Director. Conduct calibration session and review editor usage patterns.
M-04	Control Adherence Rate (R-04, R-05)	Percentage of submissions with a fully completed Human Review Checklist	< 100% (HIGH)	Daily non-compliance reporting. Escalate to Editorial Director. User access may be suspended pending retraining.
M-05	Audit Log Integrity (R-05)	Percentage of missing or incomplete records identified during audit log reconciliation	> 0.1% (HIGH)	Notify Technical Lead and IT. Initiate remediation under 48-hour SLA and conduct incident review.
M-06	Copyright Flag Rate (R-03)	Count of outputs flagged by editors for potential verbatim	> 0 instances (CRITICAL)	Immediate P1 incident. Escalate to Legal Compliance for

		copyrighted text		review, containment, and resolution.
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4. Reporting & Review Cadence

Report/Meeting	Frequency	Audience	Purpose
Model Health Dashboard	Real-Time	Editorial Director, MRM Lead	Monitor system status and active alerts
Monthly Performance Review	Monthly	Editorial Director, MRM Lead	Analyze trends, investigate anomalies, validate control effectiveness
Quarterly Governance Review	Quarterly	All Stakeholders (including Legal)	Review risk posture, incidents, and model classification

5. System Requirements (High-Level)

Implementation of this monitoring framework requires:

- Centralized logging of model interactions and checklist completion
- A dashboard interface for metric visualization
- An alerting mechanism capable of notifying stakeholders upon threshold breaches

Note: This specification defines governance requirements and does not prescribe specific tools or platforms.

6. Approvals

Role	Signature	Date
Model Governance Lead	_____	April 2026
Model Risk Management (MRM) Lead	_____	April 2026
Editorial Director (Business Owner)	_____	April 2026